Mathematical Foundations of Political Methodology

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- Sequence
- Limits of Sequence
- Cauchy Sequence
- Limits of Functions
- Continuity

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Sequence

Sequence can be irregular, an example of such sequence is:

$$a_1 = 1$$

$$a_2 = 8$$

$$a_3 = 6$$

$$a_4 = ...$$

Sequence can be regular, an example of such sequence is:

$$x_1 = 2$$

$$x_2 = 4$$

$$x_3 = 6$$

$$x_4 = ...$$

Sequence

Definition

A *sequence* *of real numbers is a function, u, mapping $\mathbb{N}$ to $\mathbb{R}$*

- We will be dealing with infinite sequences
- Written $u(1), u(2), u(3), ...$, or $(u_1, u_2, u_3, ...)$
- The sequence is written $\{u_n\}$
- Example of a sequence

$$u_n = \frac{3n^2}{n^3 + 9}$$

which gives the sequence

$$\frac{3}{10}, \frac{12}{17}, \frac{48}{73}, \ldots$$

Arithmetic Sequences

Definition

An arithmetic sequence is a sequence $\{u_n\}$ *in which*

$$u_{n+1} - u_n$$

is constant for all n

- Equivalent to saying that for some numbers a, d, the sequence looks like

$$a, a + d, a + 2d, a + 3d, \ldots$$

  or

$$u_n = a + (n - 1)d$$

- Is this sequence arithmetic? (If so, what are a, d?)

$$u_n = 14 - 5n$$

Geometric Sequences

Definition

Geometric sequences: Each term is obtained by multiplying previous term by same number.

- Equivalent to saying that for some numbers a, x the sequence looks like

$$a, ax, ax^2, ax^3, ...$$

  or

$$u_n = ax^{n-1}$$

- What are a, x for this geometric sequence?

$$u_n = 5^{-n}$$

Limits of Sequence: Motivation

Now, we want to ask a question, as n goes further and further, that is, n goes to infinity, what is the behavior of the sequence?

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Limits of sequences

Definition

A sequence $\{u_n\}$ tends to the limit z if for any positive real number ϵ there is a positive integer, N, such that:

$$|u_n - z| < \epsilon \textit{ for all } n > N.$$

Also written as

$$\lim u_n = z$$

$$\lim_{n \to \infty} u_n = z$$

$$u_n \to z \textit{ as } n \to \infty$$

Limits of Sequences

- Some Remarks
  - If a sequence converges, it converges to one number. We call that A
  - $\epsilon > 0$ is some arbitrary real-valued number.
  - The N will depend upon ϵ
  - Implies the sequence never gets further than ϵ away from A

Limits of Sequences

Theorem

$\left\{\frac{1}{n}\right\}$ converges to 0

Proof.

We need to show that for ϵ there is some N_ϵ such that, for all $n \geq N_\epsilon$ $|\frac{1}{n} - 0| < \epsilon$. Without loss of generality (WLOG) select an ϵ. Then,

$$\begin{aligned} |\frac{1}{N_\epsilon} - 0| &< \epsilon \\ \frac{1}{N_\epsilon} &< \epsilon \\ \frac{1}{\epsilon} &< N_\epsilon \end{aligned}$$

For each epsilon, then, any $N_\epsilon > \frac{1}{\epsilon}$ will suffice. □

Cauchy Sequence: Motivation

A fun fact, how do we pronounce Cauchy, KohShee.

In previous example, we show $\frac{1}{n}$ converges to 0 because we know its limit. However, a natural follow-up question to ask is that can we show that a sequence has limit, but without knowing the limit ex ante.

Cauchy sequence is developed for this purpose.

Divergent Sequence

Definition

A sequence that tends to a limit is called convergent; a sequence that doesn't is called divergent.

- A sequence that tends to infinity (or negative infinity) is divergent, but a sequence doesn't have to tend to infinity to be divergent. (What's an example of a divergent sequence that doesn't tend to infinity?)

Cauchy Sequence

Definition

$\{x_n\}$ is Cauchy if $\forall\epsilon > 0, \exists N \in \mathbb{N} : |x_m - x_n| < \epsilon$ when $m, n > N$

Theorem

In R, $\{x_n\}$ is convergent if and only if it's Cauchy.

Apply Cauchy Criterion

We can show the sequence defined by $x_n = n$ is not convergent:

$|x_n - x_m| = |n - m| \geq 1$ for all n, m. Thus, for $\epsilon = 1$ there is no N such that $|x_n - x_m| < \epsilon$ whenever $n, m > N$. Therefore $\{x_n\}$ is not Cauchy, and does not tend to a limit.

Working with Limits

- Fact 1: For a real number x, $\lim_{n\to\infty} x^n = 0$ if and only if $-1 < x < 1$.
- Fact 2: For a real number b, $\lim_{n\to\infty} n^b = 0$ if and only if $b < 0$.

Working with Limits: Evaluate Dominating Term

What is the limit of

$$u_n = \frac{(-1)^n (n+1)^2}{n^4}$$

Hint: We can start by expanding the numerator.

What would happen if we replaced $(n+1)^2$ with $(n+1)^4$?
Don't do whole calculation!

Another practice, though easier one, as $n \to \infty$, what's the limit of

$$u_n = \frac{n+1}{2n}$$

Series

- The series defined by a sequence $\{u_n\}$ is the sequence $\{s_n\}$ with

$$s_1 = u_1, s_2 = u_1 + u_2, s_3 = u_1 + u_2 + u_3, ...$$

  and more generally

$$s_n = \sum_{r=1}^{n} u_r$$

- Thus, series is a special type of sequence, it sums the terms in the sequence.

Formula for Summing Algebraic Sequence

We are interested in the sum of first n terms in arithmetic sequence. Here is the derivation:

First, the natural numbers

$$s_n = 1 + 2 + ... + (n-1) + n$$

$$s_n = n + (n-1) + ... + 2 + 1$$

Adding these together we get

$$2s_n = (1+n) + (1+n) + ... + (1+n) = n(1+n)$$

The first n terms of the natural numbers sum to

$$s_n = \frac{1}{2}n(1+n)$$

Formula for Summing Algebraic Sequence

Recall that a general arithmetic sequence is $u_n = a + (n-1)d$

We can use the same trick to find the sum of the first n terms of the sequence:

$$u_n = a + (a+d) + (a+2d) + ... + (a+(n-2)d) + (a+(n-1)d)$$

$$u_n = (a+(n-1)d) + (a+(n-2)d) + ... + (a+2d) + (a+d) + a$$

$$2u_n = n\left(2a + (n-1)d\right)$$

$$u_n = an + n\left(\frac{(n-1)d}{2}\right)$$

Summing Geometric Sequences

Summing geometric sequences: $s_n = a + ax + ax^2 + ... + ax^{n-1}$

Multiplying by x we get

$$xs_n = ax + ax^2 + ... + ax^n$$

Let

$$t = ax + ax^2 + ... + ax^{n-1}$$

Then $s_n = a + t$ and $xs_n = t + ax^n$

$$s_n - xs_n = a - ax^n$$

so

$$s_n(1 - x) = a - ax^n,$$

and finally

$$s_n = a\left(\frac{1 - x^n}{1 - x}\right)$$

An Example for Summing Geometric Sequence

Now suppose you save 1000 dollars in Citibank at year 0 (that is, the beginning year). After year 0, you will save 1000 dollars at Citibank every year. And the annual (compound) interest rate is 5 percent.

Then, a natural question will be: at year n, how much money do you make.

For the money you saved at year 0, you will get: $1000 * (1 + 0.05)^n$;

For the money you saved at year 1, you will get: $1000 * (1 + 0.05)^{n-1}$;

For the money you saved at year 2, you will get: $1000 * (1 + 0.05)^{n-2}$;

For the money you saved at year t, you will get: $1000 * (1 + 0.05)^{n-t}$;

Sum them together will give you the money you make:
$1000 * (1 + 0.05)^n + 1000 * (1 + 0.05)^{n-1} + ... + 1000 * (1 + 0.05)^1$

Convergence of Geometric Series

We will often care about the limit of a series as $n \to \infty$

For a geometric series,

$$s_n = a\left(\frac{1-x^n}{1-x}\right)$$

We know that $x^n \to 0$ as $n \to \infty$ if and only if $|x| < 1$

If $|x| < 1$ then $\lim_{n\to\infty} s_n = a\left(\frac{1-0}{1-x}\right) = \frac{a}{1-x}$

Divergence of Geometric Series

Recall that $s_n = \sum_{n=1}^{\infty} k_n$.

Fact (*term test* for any series): If $\lim_{n\to\infty} k_n \neq 0$ then $\sum_{n=1}^{\infty} k_n$ diverges

For a geometric series, $k_n = ax^{n-1}$

We know that $x^n \to 0$ as $n \to \infty$ <u>if and only if</u> $|x| < 1$

Therefore, a geometric series diverges whenever $|x| \geq 1$

Example in Multiperiod Utility Function

Suppose that I plan to consume x units of cake in each of an infinite number of time periods, $t = 1, 2, 3, ...$

I discount the utility I'll receive from future consumption by a term $0 < \delta < 1$, so that I value today's consumption fully as $u(x)$, tomorrow's as $\delta u(x)$, the day after's as $\delta^2 u(x)$...

Now, let's see a concrete example, let's set $\delta = 0.9$. Then, tomorrow's utility is: $0.9 * u(x)$, the day after's as $0.9^2 * u(x)$ ...

Example in Multiperiod Utility Function

My expected utility over time is to sum my utilities together:

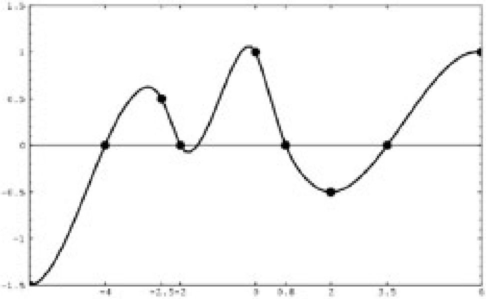
$$\delta^0 u(x) + \delta^1 u(x) + \delta^2 u(x) + \delta^3 u(x) + ...$$

We can write this summation in a compact way:

$$\sum_{t=0}^{\infty} \delta^t u(x) = \frac{u(x)}{1-\delta}$$

"Cake eating problem": How to allocate a fixed unit of x over this infinite stream *today* in order to maximize my lifetime utility

Limits of Functions: Motivation



We care about the limits of function because we care about how does function behave at a point. Essentially, this is the most important aspect in calculus (more broadly to say, real analysis): the study of limit.

And the more calculus and real analysis you learn, you will realize the importance of limits.

Limits of Functions

Definition

Suppose $f : R \to R$. We say that f has a limit L at x_0 if, for each $\epsilon > 0$, there is a $\delta > 0$ such that $|x - x_0| < \delta$ implies that $|f(x) - L| < \epsilon$.

- Limits are about the behavior of functions at points. Here x_0.
- As with sequences, we let ϵ define an error rate
- δ defines an area around x_0 where $f(x)$ is going to be within our error rate

Limits of Functions: Example

Theorem

The function $f(x) = \frac{x^2-1}{x-1}$ has a limit of 2 at $x_0 = 1$.

Proof.

For all $x \neq 1$,

$$\begin{aligned} \frac{x^2-1}{x-1} &= \frac{(x+1)(x-1)}{x-1} \\ &= x+1 \end{aligned}$$

Choose $\epsilon > 0$ and set $x_0 = 1$. Then, we're looking for δ_ϵ such that

$$|x-1| < \delta_\epsilon \quad \text{implies} \quad |(x+1)-2| < \epsilon$$

Again, if $\delta_\epsilon = \epsilon$, then this is satisfied. □

Remarks on Limits of Functions

$\lim_{x \to x_o} f(x)$ might or might not equal $f(x_o)$

- It could be *different than* $f(x_o)$

$$f(x) = \begin{cases} +1 & \text{if } x \neq 4 \\ -1 & \text{if } x = 4 \end{cases}$$

- It could be *undefined*

$$f(x) = \begin{cases} +1 & \text{if } x < 0 \\ -1 & \text{if } x \geq 0 \end{cases}$$

Algebra of Limits

Theorem

Suppose $f : \Re \rightarrow \Re$ *and* $g : \Re \rightarrow \Re$ *with limits* A *and* B *at* x_0*. Then,*

$$\begin{aligned} i.)\ \lim_{x \rightarrow x_0} (f(x) + g(x)) &= \lim_{x \rightarrow x_0} f(x) + \lim_{x \rightarrow x_0} g(x) = A + B \\ ii.)\ \lim_{x \rightarrow x_0} f(x) g(x) &= \lim_{x \rightarrow x_0} f(x) \lim_{x \rightarrow x_0} g(x) = AB \end{aligned}$$

Suppose $g(x) \neq 0$ *for all* $x \in \Re$ *and* $B \neq 0$ *then* $\frac{f(x)}{g(x)}$ *has a limit at* x_0 *and*

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow x_0} f(x)}{\lim_{x \rightarrow x_0} g(x)} = \frac{A}{B}$$

Continuity

Definition

Suppose $f : \Re \rightarrow \Re$ and consider $x_0 \in \Re$. We will say f is continuous at x_0 if for each $\epsilon > 0$ there is a $\delta > 0$ such that if,

$$\begin{aligned} |x - x_0| &< \delta \text{ for all } x \in \Re \text{ then} \\ |f(x) - f(x_0)| &< \epsilon \end{aligned}$$

- Previously $f(x_0)$ was replaced with L.
- Now: $f(x)$ has to converge on itself at x_0.
- Continuity is more restrictive than limit